

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and maintained **data pipelines and ETL workflows** using **Python, Apache Spark, and Apache Airflow** orchestration, processing **500M+ daily transactions** across banking and payments infrastructure with **99.9% accuracy** and sub-second latency for real-time reporting.
- Built **data warehouse schemas** and optimized **SQL queries** across **PostgreSQL, MySQL, and Oracle** databases, reducing query execution time by **45%** through indexing strategies, partitioning, and cost-based optimizer tuning for cross-system analytics.
- Developed **machine learning models** using **scikit-learn, pandas, and NumPy** for fraud detection, transaction anomaly scoring, and predictive analytics in payments processing; achieved **94% precision** on unsupervised clustering of transaction patterns.
- Engineered **data integration pipelines** connecting enterprise systems including **ERP modules, REST APIs, and messaging platforms** (SWIFT MT/MX, ISO 20022); automated **regulatory compliance reporting** (GDPR, PSD2, SEPA) and reconciliation workflows reducing manual effort by **70%**.
- Architected **cloud-native data solutions** on **AWS (EC2, S3, Lambda, Glue)** with **Docker containerization** and **CI/CD deployment via Jenkins and Git**; implemented **monitoring and alerting** using **CloudWatch and Dynatrace** achieving **SLA compliance >99%** for production data workloads.
- Created interactive **Power BI and Tableau dashboards** with **DAX measures, Power Query ETL**, and data modelling for executive reporting; published self-service analytics reducing analytics request turnaround by **60%** and enabling **A/B testing** and statistical hypothesis validation.

SKILLS

Data Engineering & Pipelines – Apache Spark, Apache Airflow, ETL, data pipelines, ELT, workflow orchestration, Python scripting, batch and streaming data processing.

Data Warehousing & Databases – SQL, PostgreSQL, MySQL, Oracle, data warehouse design, schema optimization, query tuning, dimensional modelling, data marts.

Machine Learning & Analytics – scikit-learn, pandas, NumPy, supervised learning, unsupervised learning, regression, clustering, classification, feature engineering, model evaluation, statistical analysis.

Cloud & Infrastructure – AWS (EC2, S3, Lambda, Glue, Athena), Docker, containerization, CI/CD, Jenkins, Git, Linux, infrastructure automation.

Business Intelligence & Visualization – Power BI, Tableau, DAX, Power Query, data modelling, interactive dashboards, KPI reporting, data storytelling.

Domain & Compliance – SWIFT messaging (MT/MX), ISO 20022 message flows, SEPA, payments processing, ERP integration, GDPR, PSD2, Open Banking, regulatory compliance, financial services.

Monitoring & Observability – Splunk, Dynatrace, CloudWatch, application performance monitoring, incident response, SLA management, production support, alerting.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

PROJECTS

Real-Time Payments & Transaction Analytics Platform (HCL Technologies)

- Engineered end-to-end **Apache Spark-based data pipeline** ingesting **SWIFT MT/MX and ISO 20022 message streams** from banking payment infrastructure, validating **500M+ daily transactions** against regulatory compliance rules (GDPR, PSD2, SEPA) with **zero false negatives**.
- Built **data warehouse** using **PostgreSQL and Snowflake** with **dimensional schemas** optimized for **Power BI dashboards**; implemented **DAX measures and Power Query transformations** enabling real-time visibility into transaction flows, SLA metrics, and reconciliation status across banking operations.
- Deployed **machine learning anomaly detection models** using **scikit-learn and unsupervised clustering** to identify suspicious transaction patterns; reduced fraud investigation time by **55%** and improved detection precision to **96%** for cross-border payment monitoring.
- Orchestrated **Apache Airflow DAGs** automating **ETL schedules, data quality checks, and alerting**; integrated with **AWS Lambda and CloudWatch** for serverless job execution, reducing infrastructure costs by **40%** while maintaining **99.95% pipeline uptime**.

AI-Powered Resume Generator with LLM Integration (Personal Project)

- Built full-stack **AI workflow automation system** leveraging **Claude API (Anthropic), prompt engineering, and constraint guardrails** to generate job-description-tailored resumes at scale using structured JSON payloads and multi-turn LLM orchestration.
- Implemented **data pipeline architecture** combining **web scraping (Apify), job description ingestion, Claude API integration, and PDF generation (reportlab)**; processed **1000+ job listings monthly** with **98% accuracy** in content personalization.
- Deployed CI/CD automation using **GitHub Actions** for continuous data ingestion and integration testing; hosted **Vercel-deployed web frontend** with **React** for real-time resume generation; demonstrated proficiency in **prompt engineering, API integration, and full-stack data automation**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)